

Sailaubay Mariya 25B031793

Practice: 14-15:00

1. Identify Entities

- Airport
- Airline
- Flight
- Passenger
- Booking
- BoardingPass
- Baggage
- BaggageCheck
- SecurityCheck

2. Attributes

PK = primary key. FK = foreign key. Empty cells mean that the attribute may be left blank.

Airport

Attribute	Type	Key / rule
airport_id	INT	PK
airport_name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	
city	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Airline

Attribute	Type	Key / rule
airline_id	INT	PK
airline_code	VARCHAR(10)	UNIQUE, NOT NULL
name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Attribute	Type	Key / rule
booking_id	INT	PK
flight_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL
status	VARCHAR(30)	NOT NULL
booking_platform	VARCHAR(50)	
ticket_price	DECIMAL(10,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL

updated_at	TIMESTAMP	NOT NULL
------------	-----------	----------

Flight

Attribute	Type	Key / rule
flight_id	INT	PK
departing_gate	VARCHAR(10)	
arriving_gate	VARCHAR(10)	
airline_id	INT	FK, NOT NULL
departure_airport_id	INT	FK, NOT NULL
arrival_airport_id	INT	FK, NOT NULL
scheduled_departure	TIMESTAMP	NOT NULL
scheduled_arrival	TIMESTAMP	NOT NULL
actual_departure	TIMESTAMP	
actual_arrival	TIMESTAMP	
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Passenger

Attribute	Type	Key / rule
passenger_id	INT	PK
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
gender	VARCHAR(20)	
date_of_birth	DATE	
country_of_citizenship	VARCHAR(100)	
country_of_residence	VARCHAR(100)	
passport_number	VARCHAR(50)	UNIQUE
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Booking BoardingPass

Attribute	Type	Key / rule
boarding_pass_id	INT	PK
booking_id	INT	FK, UNIQUE, NOT NULL
seat	VARCHAR(10)	
boarding_time	TIMESTAMP	
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Baggage

Attribute	Type	Key / rule
baggage_id	INT	PK
booking_id	INT	FK, NOT NULL
weight_kg	DECIMAL(6,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

BaggageCheck

Attribute	Type	Key / rule
baggage_check_id	INT	PK
booking_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL
check_result	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

SecurityCheck

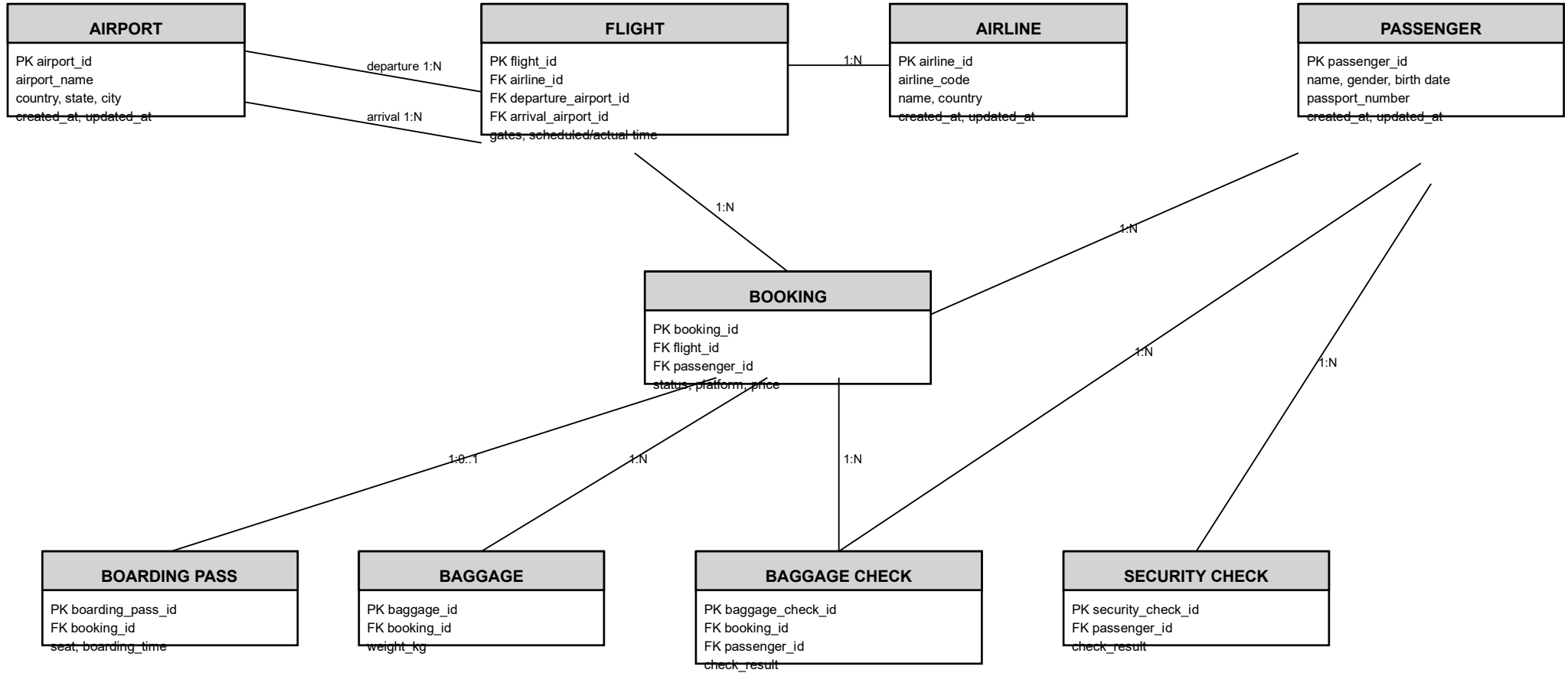
Attribute	Type	Key / rule
security_check_id	INT	PK
passenger_id	INT	FK, NOT NULL
check_result	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3. The following relations are used in the diagram. The first side is one record; the second side can have many records.

Relationship	Type	Meaning
Airline - Flight	1:N	One airline has many flights.
Airport - Flight (departure)	1:N	One airport can be the departure airport for many flights.
Airport - Flight (arrival)	1:N	One airport can be the arrival airport for many flights.
Passenger - Booking	1:N	One passenger can have many bookings.
Flight - Booking	1:N	One flight can have many bookings.
Booking - BoardingPass	1:0..1	A booking can have one boarding pass or no pass yet.
Booking - Baggage	1:N	One booking can have many baggage items.
Booking - BaggageCheck	1:N	One booking can have many baggage check records.
Passenger - BaggageCheck	1:N	One passenger can have many baggage check records.
Passenger - SecurityCheck	1:N	One passenger can have many security checks.

4. ER Diagram

The lines use 1:N unless the line says 1:0..1.



Key: PK = primary key; FK = foreign key; 1:N = one-to-many; 1:0..1 = one optional boarding pass.

Two Airport-Flight lines are shown because a flight has a departure airport and an arrival airport.

5. Short Textual Description

The database is for an international airport. It stores airports, airlines, flights, passengers and bookings. A flight belongs to one airline. It also has one departure airport and one arrival airport.

A passenger can make many bookings. A booking is for one passenger and one flight. A booking can have a boarding pass and can include baggage. The system also stores baggage check results and passenger security check results.

6. Legend

PK means primary key - a unique id of a record. FK means foreign key - a link to another table. NOT NULL means the value is required. UNIQUE means that the value cannot repeat. The signs 1:N and 1:0..1 show the relationship between tables.

Note: The last paragraph of the task also names accounts, booking legs, aircraft and frequent flyer programs. They are not included because no attributes are given for them.

